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Question #1:

For the given data points (1, 3), (3, -6), and (5, -2), which formula is useful in finding the interpolating polynomial?

Options:

A) Newton's forward difference formula

B) Newton's backward difference formula

C) Lagrange's interpolation formula

Correct option: C) Lagrange's interpolation formula

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Question #2:

If f(x) = 2x^3 - 5x^2 + 9x - 6, then its derivative is zero for all x.

Options:

A) 1

B) 6

C) 3

D) 4

Correct option: B) 6

Question #3:

For the given data points (4, 2.2), (8, 3.5), and (12, 4.1), what will be the divided difference table?

Options:

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A) Equally spaced

B) Unequally spaced

C) Union of equally spaced and unequally spaced intervals

D) Any arbitrary partition will work

Correct option: B) Unequally spaced

Question #4:

Richardson extrapolation is a method also known as:

Options:

A) Series acceleration method

B) Sequence acceleration method

Correct option: B) Sequence acceleration method

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Question #5:

In Lagrange's interpolation, for n values of y corresponding to n values of x, we can represent the function f(x) by a polynomial of degree:

Options:

A) n + 1

B) n - 1

C) n + 2

D) n

Correct option: B) n - 1

Question #6:

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Which of the following is the Richardson's Extrapolation limit: F3(h/8), provided that F2(h/8) = F2(h/4) = 1?

Options:

A) -1

B) 1

C) 64

D) 63

Correct option: B) 1

Question #7:

For a function 'f(x) = x' with a step size of 'h=0.01', which of the following gives the 1st derivative at x = 1 using the two-point formula?

Options:

A) y'(x) = 0.1 + Some Truncation Error

B) y'(x) = -0.1 + Some Truncation Error

C) y'(x) = 1.01 + Some Truncation Error

D) y'(x) = 1 + Some Truncation Error

Correct option: D) y'(x) = 1 + Some Truncation Error

Question #8:

To take the derivative of f(x) = 2x in the interval [-3,3], which of the following partition of subintervals will be suitable?

Options:

A) Union of equally spaced and unequally spaced intervals

B) Unequally spaced

C) Equally spaced

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D) Any arbitrary partition will work

Correct option: D) Any arbitrary partition will work

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